AA 279B - Project Proposal

Colin Skinner, Kabir Cheema

October 13, 2026

Orbit Insertion to Constellation

We will investigate the construction of a satellite constellation from a single orbital insertion using a mix of low-thrust propulsion and natural perturbations to achieve the desired orbital elements for each satellite.

Our primary objective is to create a finite-thrust maneuver optimizer that weighs quantities such as ΔV, propellant consumption, or time, and potentially how different constellation configurations will affect these.

Methodology

As stated, the two primary mechanisms for satellite deployment that we will utilize are:

1. **Phasing**: By changing the orbital period of the satellite, we can achieve varying anomalies/phases in the same orbital plane.
2. J_2 **Nodal Precession**: We can exploit the Earth's oblateness to naturally shift the ascending node, achieving a precession that allows us to separate orbital planes over time, with less propellant than direct plane changing.

To simulate, we will implement a numerical propagator, and our burn planner framework will use both analytic orbital mechanics as well as shooting-based methods to determine a desirable thrust sequence and profile. Depending on computation time, we may adjust our incorporation of analytic orbital dynamics to accelerate long-duration time horizons. For example, Kepler's equation can save time on propagating two-body propagation between maneuvers, while long-term propagation can be approximated via J_2 effects.

To parameterize the search space for our burns, we will impose finite, maximum, and minimum thrust levels at more discrete, spaced-out time intervals.

Analysis

We will compare deployment strategies across:

- **Different initial orbits**: Varying starting inclination, semi-major axes, etc.
- **Constellation geometries**: Number of satellites and orbital planes
- **Propulsion capabilities**: Maximum thrust and I_{sp}
- **Thrust vs. Orbit Perturbation**: Changing the degree to which we utilize both in finding optimal burns
- **Several launch sites**: If we have time, we may consider Cape Canaveral, Vandenberg, and potentially Boca Chica or Wallops

The end goal is a framework that allows us to evaluate how a constellation can be efficiently deployed from a single orbit, along with comparisons to identify when different deployment strategies will become more or less effective.